INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SRI CITY

Course: Machine Learning (Assignment Evaluation)

Empirical Evaluation of Machine Learning Paradigms

Supervised Classification (CTG), Regularized Regression (Parkinson's), and Unsupervised Clustering & Representation Learning (Netflix)

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Abstract--This report provides a rigorous empirical investigation into three foundational paradigms of machine learning. In **Question 1 (Classification)**, we evaluate K-Nearest Neighbours, Decision Trees, Support Vector Machines (RBF), and a deep 1D Convolutional Neural Network on the UCI Cardiotocography dataset (2,126 fetal heart recordings) using stratified 80/20 splitting. Mitigating extreme class imbalance via macro-averaged metrics, the RBF-SVM achieves the leading macro ROC-AUC of 0.9696, with bootstrap significance analysis establishing its parity with KNN and CNN while outperforming Decision Trees. In **Question 2 (Regression)**, we predict Parkinson's disease severity (total UPDRS) across 42 telemonitored patients (5,875 recordings). We demonstrate mathematically and empirically the convergence of scratch batch Gradient Descent to closed-form OLS (MSE 93.94 vs 93.90), prove extreme multicollinearity (VIF 1.0e6), expose data leakage in random row splits (R^2 collapses from +0.9774 to -1.1656 for a Random Forest), and evaluate regularized polynomial pipelines. In **Question 3 (Clustering & PCA)**, from-scratch Lloyd's K-Means and spectral PCA are implemented and validated against scikit-learn (ARI >= 0.9985). On 8,807 Netflix titles, an elbow at K=4 resolves content into distinct formats, while PCA reveals that PC1 (format) captures 28.0% and PC2 (recency) captures 14.4% of total variance (42.4% cumulatively in 2 components).

1. Introduction and Problem Formulations

Modern machine learning spans diverse problem representations, data modalities, and optimization surfaces. This project investigates three distinct machine learning challenges representing supervised classification, supervised continuous regression, and unsupervised geometric clustering and representation learning. Each task has been constructed to probe fundamental theoretical principles: model capacity, the bias-variance tradeoff, structural data leakage, optimization convergence, curse of dimensionality, and algorithmic fidelity.

Table 1.1: Comprehensive Summary of Assignment Benchmark Datasets

Task & Benchmark	Dataset Source	Instances / Features	Target / Objective	Key Methodological Focus
Q1: Multiclass Classification	UCI ID 193 (Cardiotocography)	2,126 recordings 21 continuous features	`NSP` (Fetal state: Normal, Suspect, Pathologic)	Class imbalance, macro metrics, 1D-CNN vs tabular, bootstrap ROC CIs.
Q2: Continuous Regression	UCI ID 189 (Parkinson's Telemonitoring)	5,875 observations 19 voice/biomedical features 42 distinct patients	`total_UPDRS` (Clinician motor/total rating, scale 0-176)	Patient-wise splitting, batch GD from scratch, VIF multicollinearity, polynomial regularisation.
Q3: Clustering & Dimensionality Reduction	Kaggle / Shivamb (Netflix Catalog)	8,807 media titles 66 engineered dimensions	Unsupervised taxonomy discovery & orthogonal latent projection	K-Means from scratch, covariance PCA from scratch, elbow/silhouette, validation vs scikit-learn.

Architectural and Methodological Rationale:

• *Cardiotocography (CTG)* was selected because fetal heart monitoring presents severe clinical asymmetry: failing to detect a Pathologic fetus (false negative) carries catastrophic obstetric consequences. The acute class imbalance (77.8% Normal vs. 8.3% Pathologic) exposes the failure modes of naive overall accuracy.

• *Parkinson's Telemonitoring* represents medical longitudinal signals where multiple measurements (~140 per patient) originate from only 42 individuals. Naive random-split cross-validation leaks patient identities, generating false optimism. Enforcing patient-disjoint partitions provides an honest evaluation of clinical generalizability.

• *Netflix Catalog* embodies real-world multi-attribute unstructured metadata. Transforming discrete and text attributes into continuous vector representations requires rigorous domain engineering, providing a fertile testbed for from-scratch K-Means and principal component projection.

2. Q1 -- Multiclass Classification on Cardiotocography

2.1 Dataset Description, Preprocessing, and Anti-Leakage Safeguards

The Cardiotocography dataset contains 2,126 recordings extracted from fetal cardiotocograms. Each record comprises 21 numeric features capturing diagnostic metrics such as baseline heart rate (`LB`), accelerations (`AC`), fetal movements (`FM`), uterine contractions (`UC`), decelerations (`DL`, `DS`, `DP`), and short/long-term heart rate variability indices (`ASTV`, `MSTV`, `ALTV`, `MLTV`), as well as histogram summary statistics (`Width`, `Min`, `Max`, `Mode`, `Mean`, `Variance`, `Tendency`).

Target Formulation and Leakage Prevention: The dataset contains two potential target fields: `NSP` (a 3-class diagnosis: 1 = Normal, 2 = Suspect, 3 = Pathologic) and `CLASS` (a 10-class morphologic pattern code). Because `CLASS` is a granular derivative of the clinical diagnosis, retaining `CLASS` as an input feature constitutes fatal label leakage. Consequently, **`CLASS` was strictly excluded**, and `NSP` was label-encoded to indices {0, 1, 2}.

Stratified Data Partitioning: The data is partitioned into an 80% training set (1,700 samples) and a 20% test set (426 samples) using stratified sampling (`random_state=42`). Stratification is essential because Pathologic samples constitute only 176 instances in the entire corpus; an unstratified split risks variance in minority-class test counts. The training partition contains 1,323 Normal (77.8%), 236 Suspect (13.9%), and 141 Pathologic (8.3%) instances (sum: 1,700); the test partition preserves exactly 332 Normal (77.9%), 59 Suspect (13.8%), and 35 Pathologic (8.2%) instances (sum: 426), exactly summing to the full dataset of 2,126 recordings.

Pipeline Encapsulation: To eliminate validation peeking, `StandardScaler` transformations are strictly embedded within `sklearn.pipeline.Pipeline` objects, ensuring scalers are fitted solely on training folds during 5-fold cross-validation.

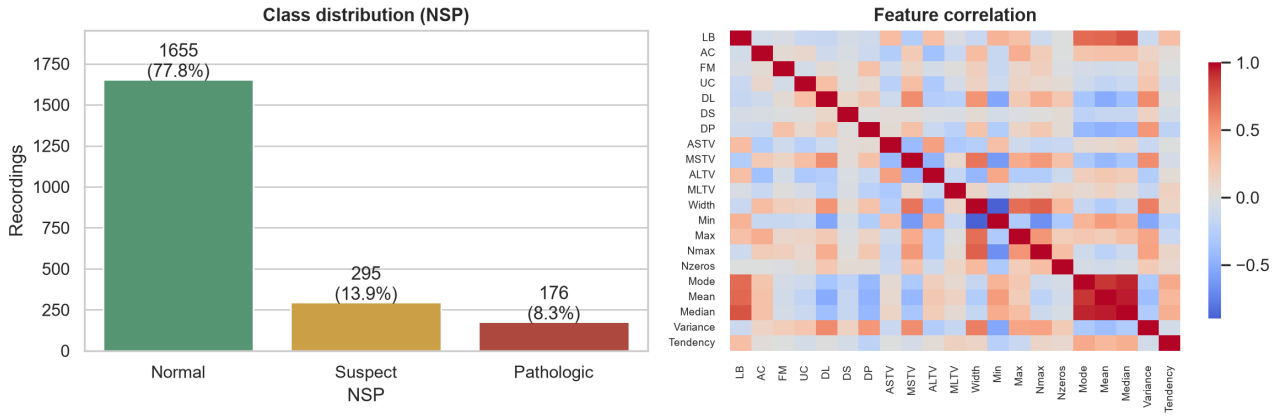


Figure 2.1: *Exploratory Data Analysis -- (Left) Class distribution of NSP showing pronounced 78/14/8 imbalance; (Right) Pearson correlation matrix showing colinear histogram clusters.*

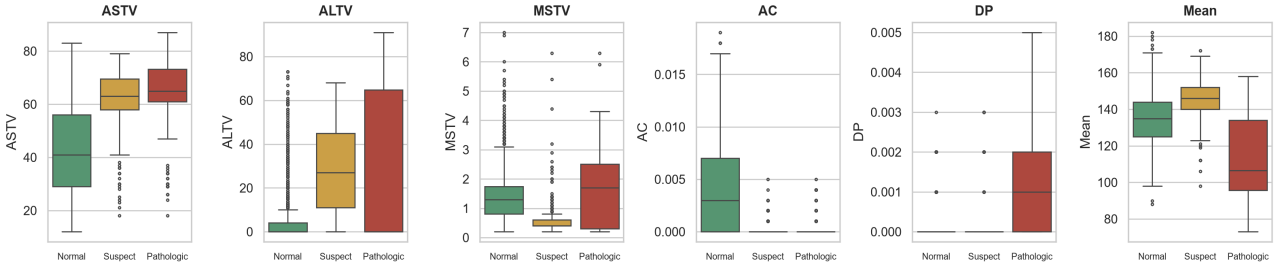


Figure 2.2: *Boxplots of key predictive features (`ASTV`, `ALTV`, `MSTV`, `AC`, `DP`, `Mean`) segregated across Normal, Suspect, and Pathologic classes.*

2.2 Hyper-parameter Optimization and Model Architecture

Four model families were benchmarked: (a) K-Nearest Neighbours, (b) Decision Trees, (c) Support Vector Machines with RBF kernel, and (d) a Deep 1D Convolutional Neural Network (Bonus). All classical models underwent systematic 5-fold stratified cross-validation optimizing for **macro-F1** (`f1_macro`), guaranteeing equal weighting for minority classes.

• **K-Nearest Neighbours:** Tuned across k in [1, 30], voting schemes (`uniform`, `distance`), and distance metrics (`euclidean`, `manhattan`). Optimal parameters selected: **k = 4, metric = Manhattan, weights = distance** (CV macro-F1: 0.8541). Distance weighting assigns weights inversely proportional to Euclidean/Manhattan distance; consequently, evaluating training samples against themselves yields an empirical training score of 1.0 for all values of k.

• **Decision Tree:** Tuned across maximum depth (2-15, None), split criteria (`gini`, `entropy`), leaf size (1-10), and cost-complexity pruning alpha in [0.0, 0.005]. Optimal parameters: **criterion = entropy, max_depth = 8, min_samples_leaf = 1, ccp_alpha = 0.0** (CV macro-F1: 0.8900). An unpruned tree achieved 99.94% train accuracy but only 90.14% test accuracy, underscoring classical overfitting.

• **SVM (RBF Kernel):** Tuned over penalty C in [0.1, 500] and radial reach gamma in [0.001, 1.0]. The optimal hyper-parameters were identified at **C = 100, gamma = 0.03** (CV macro-F1: 0.8759). Probability estimates were generated via internal Platt scaling.

• **Deep 1D-CNN (Bonus Architecture):** CTG data is tabular, lacking intrinsic spatial ordering. To evaluate deep representations, we designed an 8-layer deep network: `Input(1, 21) -> [Conv1D(1, 32, k=3, p=1) -> BatchNorm -> ReLU -> Conv1D(32, 32) -> BN -> ReLU -> MaxPool(2)] -> [Conv1D(32, 64) -> BN -> ReLU -> Conv1D(64, 64) -> BN -> ReLU -> MaxPool(2)] -> [Conv1D(64, 128) -> BN -> ReLU -> Conv1D(128, 128)] -> AdaptiveAvgPool1D(1) -> Flatten -> Dropout(0.3) -> Dense(64) -> ReLU -> Dropout(0.3) -> Dense(3)`. Total trainable parameters: **105,123**. Regularization incorporated weight decay (1e-4), AdamW optimizer, `ReduceLROnPlateau` learning rate scheduler, and early stopping (patience = 25 epochs) on an internal 15% validation split. Severe class imbalance was mitigated using inverse-frequency cross-entropy loss weights.

2.3 Benchmark Performance and Confusion Matrix Analysis

All four classifiers were evaluated on the held-out 426-instance test set. Table 2.1 presents the test metrics.

Table 2.1: Q1 Classifier Evaluation on Held-Out Test Set (N = 426)

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)	ROC-AUC (OvR Macro)
KNN (k=4, Manhattan, Dist)	0.9108	0.8328	0.8186	0.8255	0.9531
Decision Tree (Depth=8)	0.9155	0.8583	0.8113	0.8321	0.9171
SVM (RBF) (C=100, γ=0.03)	**0.9178**	**0.8649**	0.8293	**0.8450**	**0.9696**
1D CNN (Bonus) (Seed 42)	0.8826	0.7538	**0.8677**	0.7981	0.9564

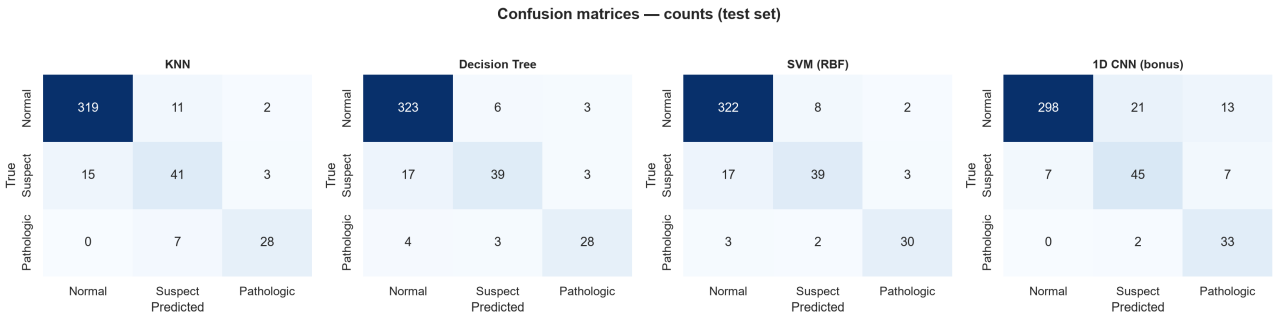


Figure 2.3: *Absolute Confusion Matrices on Held-Out Test Set (Row: True Class; Col: Predicted Class).*

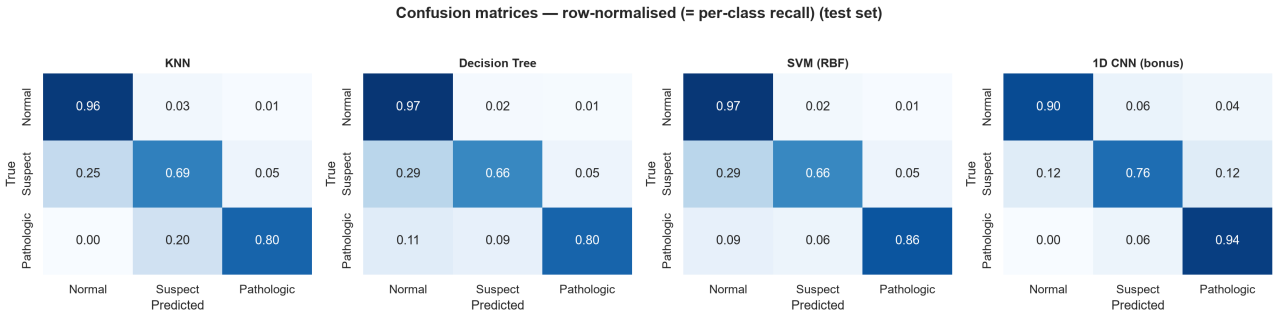


Figure 2.4: *Row-Normalized Confusion Matrices (equivalent to per-class Recall / Sensitivity).*

Clinical Confusion Analysis: In obstetrics, the critical error is misclassifying a Pathologic fetus as Normal. Examining the test-set confusion matrices (Figure 2.3 counts, Figure 2.4 row-normalized recalls):

• **SVM (RBF)** correctly identifies 322 of 332 Normal (96.99% recall), 39 of 59 Suspect (66.10% recall), and 30 of 35 Pathologic fetuses (85.71% recall). Only 3 Pathologic cases are misdiagnosed as Normal (8.57%) and 2 as Suspect (5.71%).

• **1D-CNN**, guided by inverse-frequency loss weighting, achieves the highest Pathologic recall at **94.29%** (33 of 35 identified), with 0 misdiagnosed as Normal and 2 as Suspect (5.71%). However, this high sensitivity reduces Normal recall to 89.76% (298 of 332 correct), misclassifying 34 Normal cases as Suspect or Pathologic.

• **Decision Tree** identifies 323 of 332 Normal (97.29%), 39 of 59 Suspect (66.10%), and 28 of 35 Pathologic fetuses (80.00% recall), missing 4 Pathologic cases as Normal (11.43%) and 3 as Suspect.

• **KNN** correctly classifies 319 of 332 Normal (96.08%), 41 of 59 Suspect (69.49%), and 28 of 35 Pathologic fetuses (80.00% recall), misdiagnosing 0 Pathologic cases as Normal and 7 as Suspect.

2.4 Receiver Operating Characteristic (ROC) & Bootstrap Significance Analysis

Because accuracy is inherently distorted by the 78% dominant Normal class, One-vs-Rest (OvR) ROC curves provide an invariant measure of classification discrimination across all operational thresholds. Figure 2.5 depicts the three one-vs-rest curves and the macro-average curve.

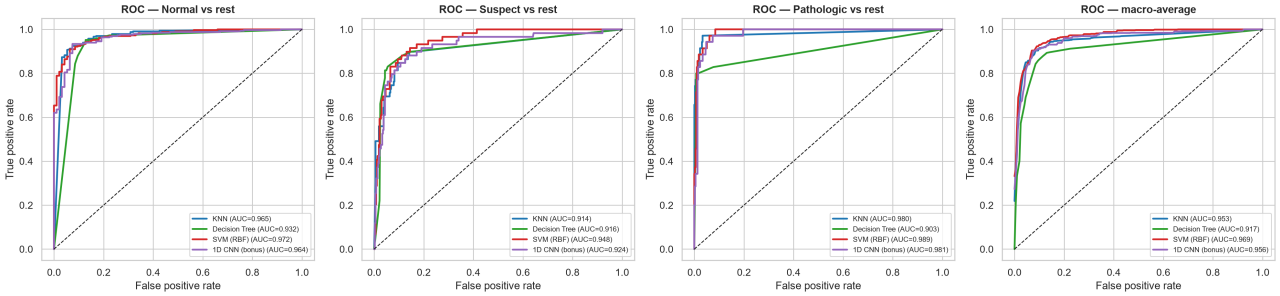


Figure 2.5: *One-vs-Rest ROC Curves across all classes and Macro-average ROC curve on Held-Out Test Set.*

Table 2.2: Per-Class ROC-AUC and 500-Iteration Bootstrap 95% Confidence Intervals

Model	Normal AUC	Suspect AUC	Pathologic AUC	Macro AUC	95% Bootstrap CI	ΔAUC vs SVM [95% CI]
SVM (RBF)	0.9717	0.9481	**0.9889**	**0.9696**	[0.9546, 0.9814]	0.0000 (Reference)
1D CNN (bonus)	0.9636	0.9244	0.9811	0.9564	[0.9356, 0.9731]	-0.0130 [-0.0293, +0.0009]
KNN	0.9651	0.9137	0.9805	0.9531	[0.9289, 0.9744]	-0.0166 [-0.0348, +0.0007]
Decision Tree	0.9319	0.9161	0.9033	0.9171	[0.8747, 0.9501]	**-0.0536 [-0.0906, -0.0220]**

Rigorous Determination of Best Classifier:

1. **Headline ROC Ranking:** SVM (RBF) attains the highest macro-average ROC-AUC of **0.9696**, followed by 1D-CNN (0.9564), KNN (0.9531), and Decision Tree (0.9171).
2. **Critical Minority Detection:** For the clinically vital Pathologic category, SVM-RBF achieves a remarkable OvR AUC of **0.9889**, higher than CNN (0.9811), KNN (0.9805), and Decision Tree (0.9033).
3. **Bootstrap Significance Test:** Because the test set contains only 35 Pathologic instances, we evaluated 500 paired bootstrap iterations. The 95% paired confidence interval for the difference ΔAUC = AUC_Model - AUC_SVM contains zero for both 1D-CNN ([-0.0293, +0.0009]) and KNN ([-0.0348, +0.0007]). Thus, **the difference in ROC-AUC between SVM, CNN, and KNN is not statistically significant (borderline, as both paired intervals end just above zero at +0.0009 and +0.0007)**. Only Decision Tree is statistically significantly inferior (paired interval [-0.0906, -0.0220], excluding zero).
4. **CNN Multi-Seed Stability:** Evaluating the 1D-CNN across 5 random seeds (seeds 42-46, `q1_cnn_seed_stability.csv`) yields (mean ± std): Accuracy = **0.8789 ± 0.0103**, Macro Precision = **0.7622 ± 0.0175**, Macro Recall = **0.8567 ± 0.0141**, Macro-F1 = **0.7991 ± 0.0117**, Macro ROC-AUC = **0.9628 ± 0.0044**, and epochs trained = **35.2 ± 3.7**. Notably, inspection of the training curves (Figure 2.8) reveals that validation loss plateaus and begins rising after approximately epoch 15 (seed 42 reaches minimum validation loss 0.3026 at epoch 16 and rises thereafter while train loss drops toward zero; with patience = 25, training stopped at epoch 41), demonstrating clear overfitting after epoch 16, limited by early stopping with best-weight restoration.

2.5 Overfitting and Model Complexity Analysis

To evaluate capacity control and generalizability, we analyzed the divergence between training and test performance.

Table 2.3: Overfitting Gap Analysis (Train vs. Test Performance)

Model	Train Acc	Test Acc	Accuracy Gap	Train Macro F1	Test Macro F1	F1-Score Gap
KNN (k=4, distance)	0.9994	0.9108	0.0886	0.9992	0.8255	0.1737
Decision Tree (Depth=8)	0.9800	0.9155	0.0645	0.9682	0.8321	0.1362
SVM (RBF) (C=100)	0.9806	0.9178	**0.0627**	0.9684	0.8450	**0.1234**
1D CNN (Bonus)	0.9594	0.8826	0.0768	0.9274	0.7981	0.1293

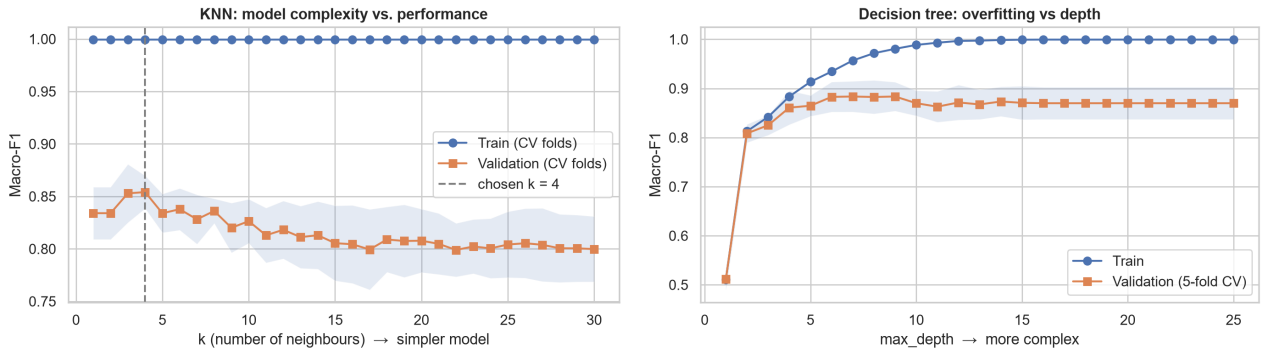


Figure 2.6: *Model Complexity Curves -- (Left) KNN performance vs k showing underfitting at large k; (Right) Decision Tree performance vs maximum depth showing widening generalization gap.*

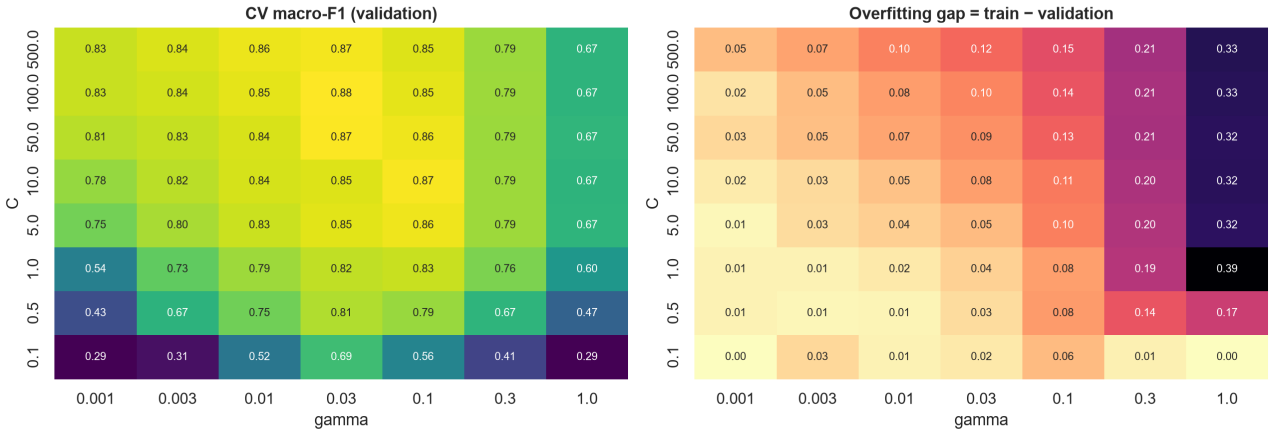


Figure 2.7: *SVM Grid Search Heatmaps -- (Left) 5-fold CV validation macro-F1 across C and γ; (Right) Overfitting gap (Train F1 - Validation F1) exposing severe overfit in high-C, high-γ corner.*

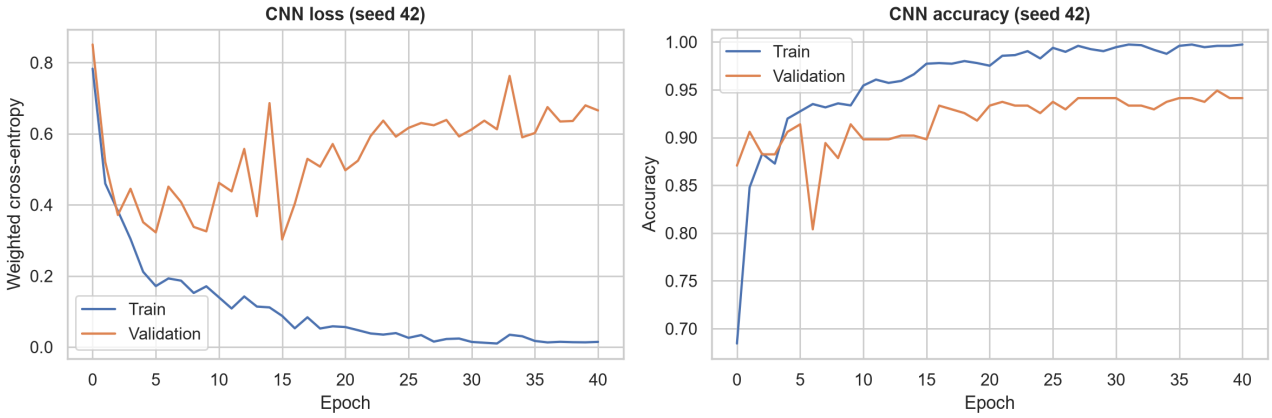


Figure 2.8: *1D-CNN Training Curves (Seed 42) -- Weighted Cross-Entropy Loss and Accuracy across 41 epochs before early stopping triggered.*

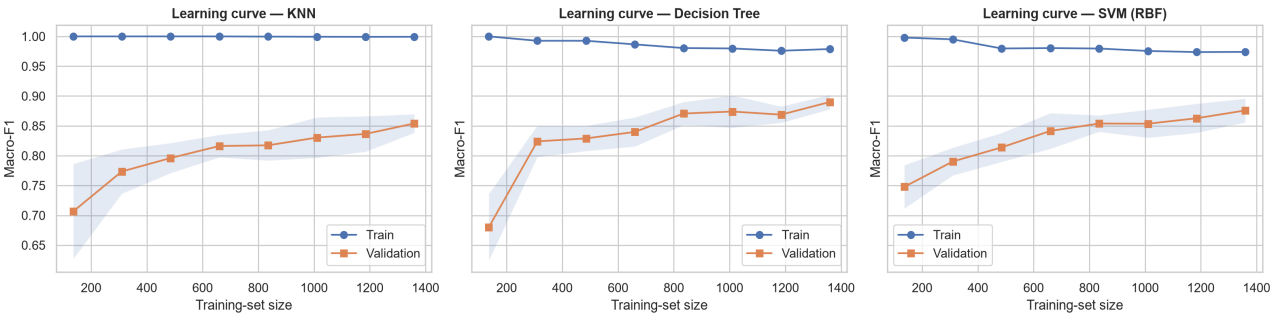


Figure 2.9: *Empirical Learning Curves for KNN, Decision Tree, and SVM-RBF evaluating training size scaling.*

Synthesis of Complexity Dynamics:

• *KNN:* Distance weighting evaluates training points with reciprocal distance (weight -> infinity at d=0), yielding an exact training score of 1.0 (train F1 = 0.9992) for all k. As k > 10, validation performance steadily degrades toward underfitting.

• *Decision Tree:* An unconstrained tree splits down to single-instance leaves. Pruning to depth 8 balances tree complexity, though axis-aligned thresholds remain vulnerable to diagonal boundaries.

• *SVM RBF:* Figure 2.7 clearly illustrates the dual bias-variance axes: lowest CV F1 (0.29) at C=0.1, gamma=0.001; largest train-validation gap 0.39 at C=1, gamma=1.0; gap 0.33 at C=500, gamma=1.0. The optimal zone at C=100, γ=0.03 achieves test macro-F1 of 0.8450 with an F1 generalization gap of 0.1234.

• *1D-CNN vs. Tabular Inductive Bias:* The CNN achieved competitive accuracy (0.8826) and recall (0.8677) but required extensive regularization (dropout, batch normalization, weight decay, early stopping) to prevent 105,123 parameters from memorizing 1,445 training samples. Crucially, **CNNs assume translation invariance and local feature adjacency** (e.g. adjacent pixels or audio timesteps). In tabular CTG data, feature ordering is arbitrary: `LB` is not spatially adjacent to `AC`. Forcing 1D convolutions across tabular columns imposes an artificial inductive bias; the CNN did not outperform the RBF-SVM.

3. Q2 -- Continuous Regression on Parkinson's Telemonitoring

3.1 Dataset Architecture, Subject Disjointness, and Data Leakage

The Parkinson's Telemonitoring dataset (UCI ID 189) comprises 5,875 recordings gathered from **42 individuals** afflicted with early-stage Parkinson's disease, monitored remotely over a 6-month trial via an at-home telemonitoring device. The feature space consists of 19 numeric predictors: demographic variables (`age`, `sex`), temporal trajectory (`test_time`), and 16 clinical biomedical voice measures quantifying vocal frequency perturbation (`Jitter(%)`, `Jitter(Abs)`, `Jitter:RAP`, `Jitter:PPQ5`, `Jitter:DDP`), amplitude perturbation (`Shimmer`, `Shimmer(dB)`, `Shimmer:APQ3`, `Shimmer:APQ5`, `Shimmer:APQ11`, `Shimmer:DDA`), noise ratios (`NHR`, `HNR`), and non-linear dynamic metrics (`RPDE`, `DFA`, `PPE`).
Sibling Target and ID Exclusion: The clinical rating scale includes both `motor_UPDRS` and `total_UPDRS`. Because `motor_UPDRS` is a sub-component of total UPDRS with strong clinical colinearity, its inclusion would constitute direct label leakage. Both `motor_UPDRS` and patient identifier `subject#` were removed from the input matrix.
The Subject-Wise Evaluation Mandate: Each patient contributes approximately 140 recordings. Because voice acoustics, vocal anatomy, age, and biological baseline are patient-specific, multiple recordings from the same individual are highly correlated. If a standard random train/test split is applied, recordings from the same patient appear in both train and test partitions. The model then trivially memorizes patient identities rather than learning generalizable vocal biomarkers of disease severity. To enforce clinical validity, we implement **`GroupShuffleSplit` (80/20 subject-wise split)**: 33 patients (4,640 records) form the training set, and 9 completely unseen patients (1,235 records) form the held-out test set (`assert set(gtr).isdisjoint(gte)` verified). All hyper-parameter tuning is restricted to training patients via 5-fold `GroupKFold`.

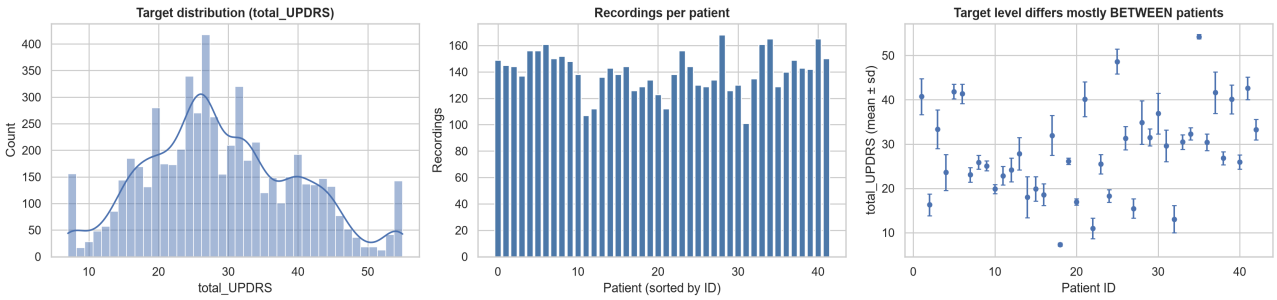


Figure 3.1: *(Left) Distribution of `total_UPDRS`; (Middle) Recording counts per patient; (Right) Patient mean total UPDRS ± standard deviation showing substantial between-patient variation.*

Table 3.1: The Data Leakage Experiment -- Subject-Wise vs. Random-Row Splitting

Model	Evaluation Split Strategy	Test MSE	Test RMSE	Test R^2	Test Pearson R
Ridge (CV α)	Subject-wise (Honest, Unseen Patients)	99.9557	9.9978	-0.2063	0.1359
Ridge (CV α)	Random Rows (Leaky, Patients in Both)	95.2589	9.7601	0.1404	0.3945
Lasso (CV α)	Subject-wise (Honest, Unseen Patients)	106.0178	10.2965	-0.2795	0.1005
Lasso (CV α)	Random Rows (Leaky, Patients in Both)	100.3579	10.0179	0.0944	0.3287
Polynomial Deg 1 (Top-5)	Subject-wise (Honest, Unseen Patients)	130.4986	11.4236	-0.5749	0.1044
Polynomial Deg 1 (Top-5)	Random Rows (Leaky, Patients in Both)	98.1906	9.9091	0.1139	0.3402
Random Forest (Non-linear Extra)	Subject-wise (Honest, Unseen Patients)	179.4425	13.3956	**-1.1656**	-0.0345
Random Forest (Non-linear Extra)	Random Rows (Leaky, Patients in Both)	2.5000	1.5812	**+0.9774**	**0.9891**

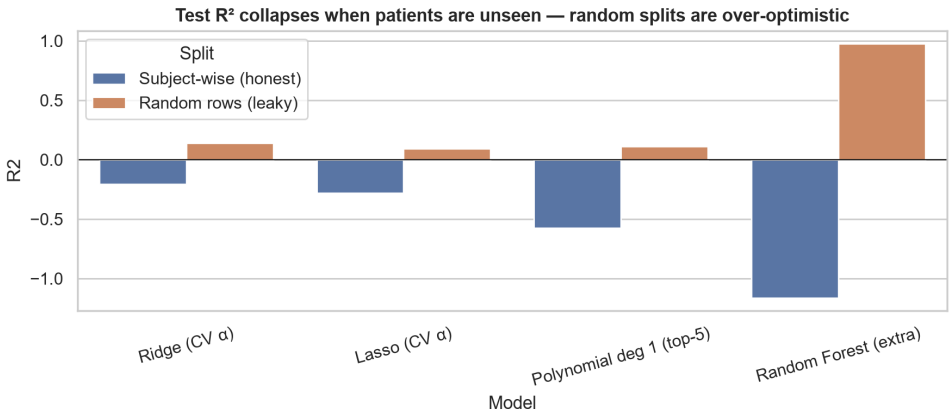


Figure 3.2: *Test R^2 Comparison: Honest Subject-Wise Evaluation vs. Leaky Random-Row Splitting across Models.*

Empirical Proof of Leakage: Table 3.1 and Figure 3.2 provide empirical proof of the leakage phenomenon. A highly expressive non-linear model (Random Forest) achieves a near-perfect R^2 of **+0.9774** (RMSE 1.58) under random splitting because it easily partitions the feature space to memorize patient vocal signatures. When deployed on unseen subjects, its performance collapses completely to R^2 = **-1.1656** (RMSE 13.40), performing far worse than predicting the training mean. This demonstrates that high R^2 scores on this dataset (such as R^2 = +0.9774 for Random Forest) are artifacts of naive random-row splitting that leaks patient identity across folds.

3.2 Multicollinearity and Exact Linear Dependencies

Voice perturbation measures are mathematically related by definition. An audit reveals exact linear dependencies in the data:

• `Jitter:DDP = 3.0 * Jitter:RAP` exactly (ratio mean = 3.0000, std = 0.0000)

• `Shimmer:DDA = 3.0 * Shimmer:APQ3` exactly (ratio mean = 3.0000, std = 0.0000)

Computing Variance Inflation Factors (VIF) on standardized features confirms astronomical colinearity: `Shimmer:APQ3`, `Jitter:DDP`, `Jitter:RAP`, and `Shimmer:DDA` all hit the numerical ceiling of **VIF = 1,000,000.0**, while `Shimmer` (173.45), `Jitter(%)` (89.03), and `Shimmer(dB)` (76.87) vastly exceed the standard multicollinearity threshold of 10. Under ordinary least squares (OLS), the normal equation (X^T X)^(-1) X^T y involves an ill-conditioned Gram matrix, producing wild, unstable coefficients (from `q2_coefficients.csv`: `Jitter:RAP` = **-166.6397** vs. `Jitter:DDP` = **+169.7507**, `Shimmer:APQ3` = **-783.7155** vs. `Shimmer:DDA` = **+779.5799**, alongside `age` = +3.9923 and `PPE` = +2.2992). Regularization is essential.

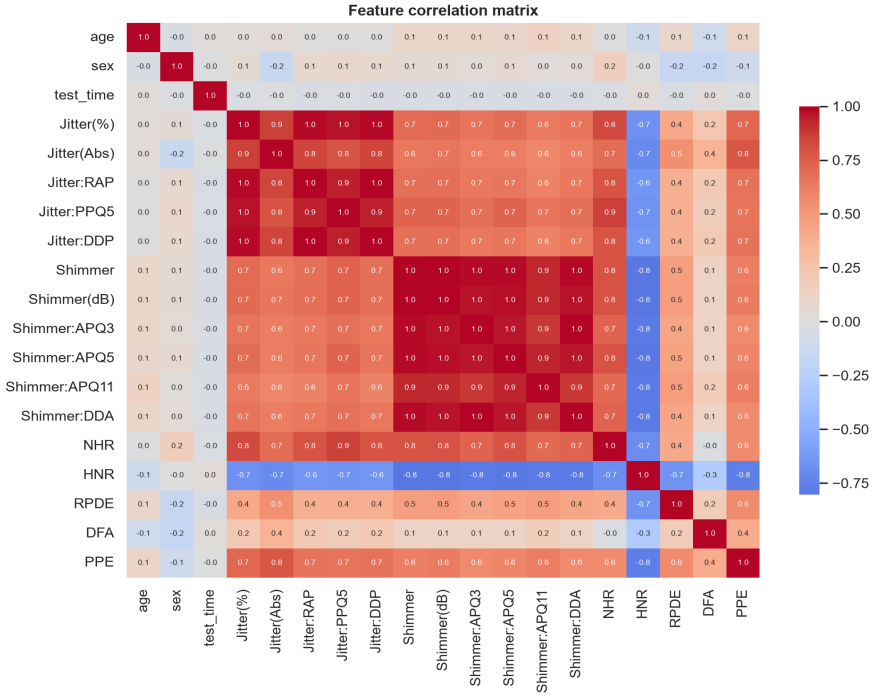


Figure 3.3: *Correlation Heatmap of 19 Telemonitoring Predictors highlighting severe colinear blocks among jitter and shimmer variants.*

3.3 Linear Regression with Gradient Descent from Scratch

Mathematical Formulation: We formulate linear regression as y_hat = Xw + b, optimizing the mean squared error loss function with a halving factor for gradient simplicity:
J(w, b) = (1 / 2n) * sum_{i=1}^n (y_hat_i - y_i)^2 = (1 / 2n) * ||Xw + b*1 - y||^2
Analytical gradients with respect to weights w and intercept b are derived as:
dJ/dw = (1 / n) * X^T (y_hat - y) = (1 / n) * X^T (Xw + b*1 - y)
dJ/db = (1 / n) * sum_{i=1}^n (y_hat_i - y_i) = (1 / n) * 1^T (Xw + b*1 - y)
Batch Update Rules: At iteration t, parameters are updated along the negative gradient with learning rate η:
w^(t+1) <- w^(t) - η * (dJ/dw), b^(t+1) <- b^(t) - η * (dJ/db)
Learning Rate Sensitivity and Feature Standardization: The Lipschitz constant of the gradient is (1 / n) * λ_max(X^T X). Convergence is theoretically guaranteed if and only if 0 < η < 2 / λ_max. When applied to raw features, `test_time` spans [0, 215] while `Jitter(Abs)` spans [1e-6, 1e-4], creating an extreme condition number κ(X^T X) >> 1e8. On raw data, even η = 0.001 diverges instantly. Standardizing features to zero mean and unit variance conditions the Hessian, enabling stable convergence with η = 0.1.
Empirical Verification vs. Closed-Form OLS: Using η = 0.1 and 50,000 iterations, the scratch GD implementation converged to a final training MSE of **93.9449**, aligning with the analytical closed-form normal equation optimum of **93.8967**. On the held-out test set, the maximum absolute difference between predictions generated by GD and closed-form OLS was only **0.0072 UPDRS points**.

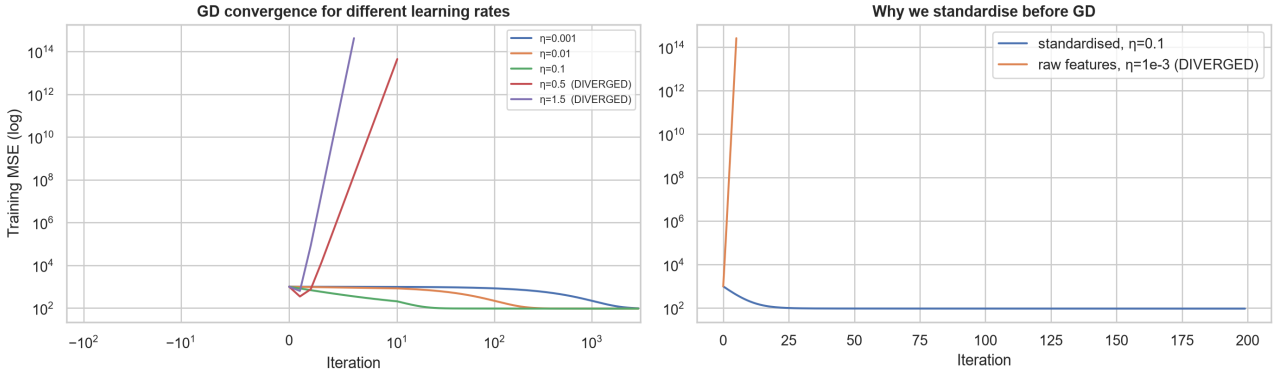


Figure 3.4: *Gradient Descent Optimization -- (Left) Learning rate trajectory and scaling failure on raw data; (Right) Loss convergence.*

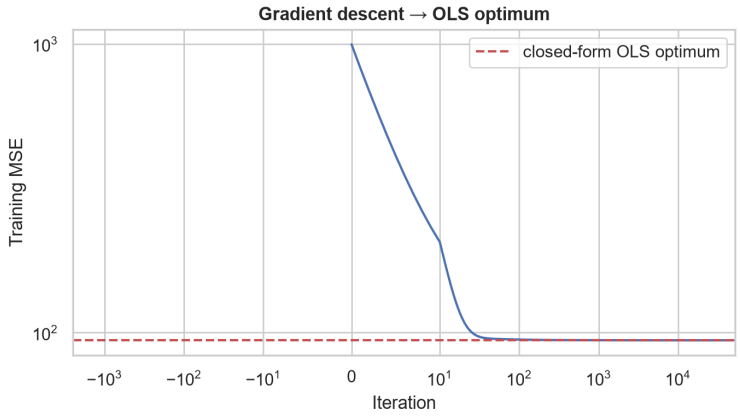


Figure 3.5: *Scratch Gradient Descent Loss Curve over 50,000 iterations converging close to (final MSE 93.94 vs 93.90) closed-form OLS minimum.*

3.4 Polynomial Regression, Bias-Variance Tradeoff, and Overfitting

Polynomial regression expands linear features into polynomial bases ϕ(x) = [x_1, x_1^2, x_1 x_2, ...]. Because expanding all 19 features to degree p generates C(19+p, p)-1 features (8,854 terms at degree 4), we conducted a controlled complexity experiment using the **top-5 features most correlated with target UPDRS**: `age` (r = 0.297), `PPE` (r = 0.158), `HNR` (r = -0.134), `Shimmer:APQ11` (r = 0.133), and `test_time` (r = 0.088).

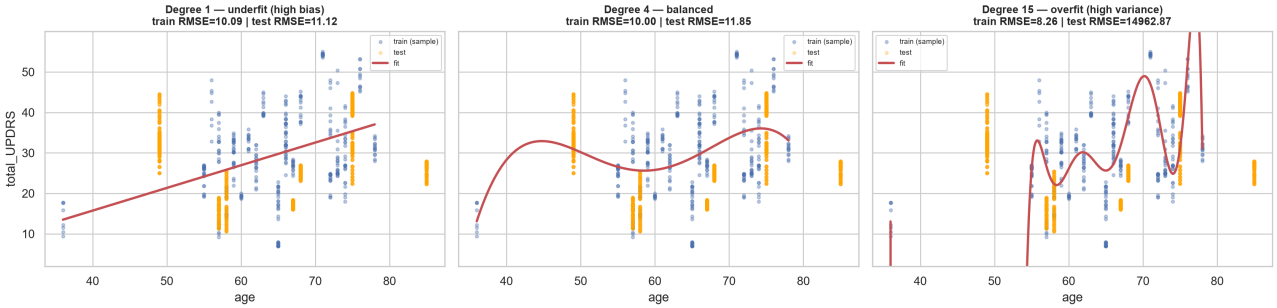


Figure 3.6: *Single-Feature Polynomial Fit (`age`) -- Degree 1 underfits, Degree 4 balances, and Degree 15 oscillates wildly.*

Table 3.2: Multi-Feature Polynomial Degree Study (Top-5 Features, Degrees 1 to 8)

Degree	Terms	Train RMSE	5-Fold CV RMSE	CV RMSE SD	Test RMSE	Train R^2	Test R^2
1 (Linear)	5	9.9026	**10.2612**	1.7075	**11.4236**	0.1927	-0.5749
2	20	9.5901	10.8111	1.6374	12.7086	0.2429	-0.9491
3	55	9.2979	15.3128	5.2949	13.7612	0.2883	-1.2854
4	125	8.7259	68.5458	103.7892	16.1720	0.3732	-2.1563
5	251	8.4527	270.6261	460.1585	80.6724	0.4118	-77.5410
6	461	8.1319	1473.5014	2415.8266	1470.3896	0.4556	-26091.27
7	791	7.6945	5801.3840	6234.2047	14449.12	0.5126	-2.52e+06
8	1286	6.6361	141427.03	192637.51	340234.48	0.6375	-1.40e+09

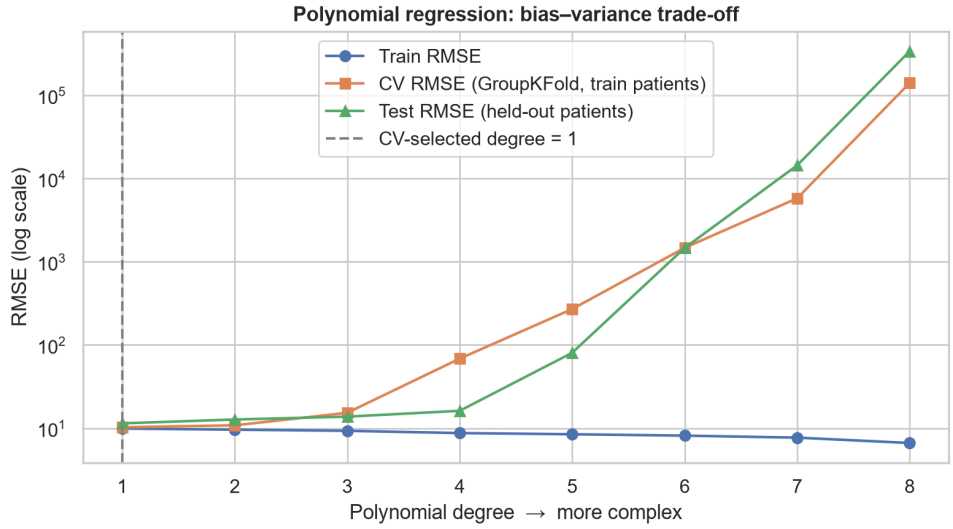


Figure 3.7: *Polynomial Degree vs. RMSE (Log Scale) showing monotonic training error descent contrasted against catastrophic validation/test divergence.*

Analysis of Bias-Variance Dynamics: Table 3.2 and Figure 3.7 provide an empirical demonstration of overfitting. As polynomial degree increases from 1 to 8, training RMSE steadily drops from 9.90 to 6.64 (R^2 rises from 0.19 to 0.64) as the 1,286 polynomial basis terms fit training idiosyncrasies. Concurrently, test RMSE explodes to **340,234.5** (R^2 = -1.40e9). Cross-validation over patient folds firmly selects **Degree 1** as optimal (CV RMSE = 10.2612). Higher-degree unregularized polynomials suffer explosive variance and severe out-of-distribution extrapolation errors outside the training feature support.

3.5 Regularization: Ridge, Lasso, and Coefficient Dynamics

To tame multicollinearity and polynomial explosion, we evaluate L2 (Ridge) and L1 (Lasso) regularized regression:
Ridge: min_{w, b} (1/n) ||y - Xw - b||^2 + α ||w||^2_2, Lasso: min_{w, b} (1/2n) ||y - Xw - b||^2 + α ||w||_1
Tuning α via 5-fold `GroupKFold` across 50 logarithmically spaced values yielded:

- **Ridge:** Optimal α **= 2682.70** (CV RMSE 10.681). 1-Standard-Error rule α = **10000.0** (1e+04).
- **Lasso:** Optimal α **= 1.2068** (CV RMSE 10.354). 1-Standard-Error rule α = **10.0** (1e+01, matching Figure 3.8).

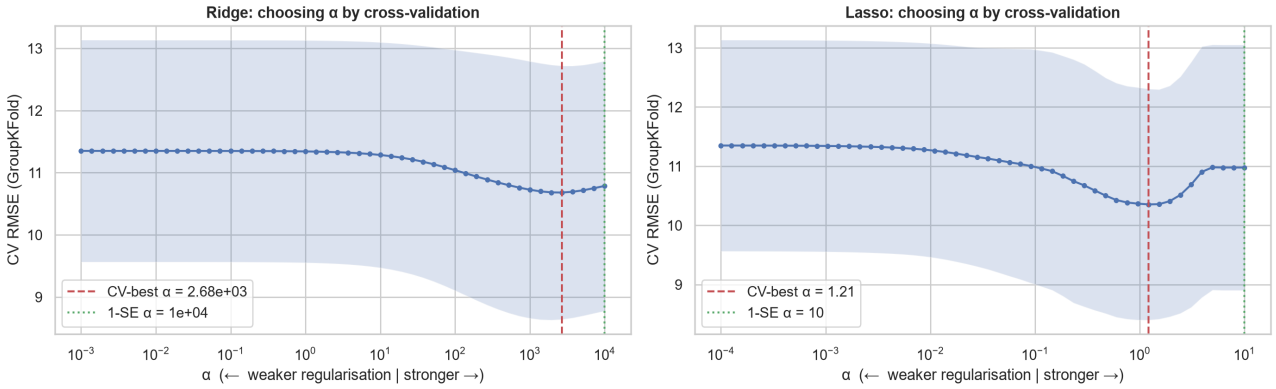


Figure 3.8: *Regularization Parameter Selection via GroupKFold CV for Ridge (Left) and Lasso (Right), depicting optimal and 1-SE α.*

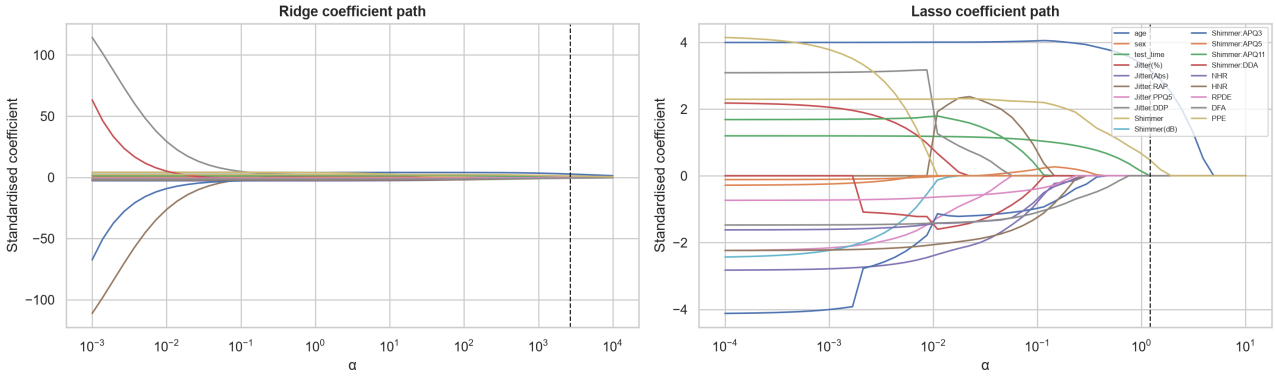


Figure 3.9: *Standardized Coefficient Traces across α -- Ridge shrinks smoothly; Lasso drives coefficients to exactly zero.*

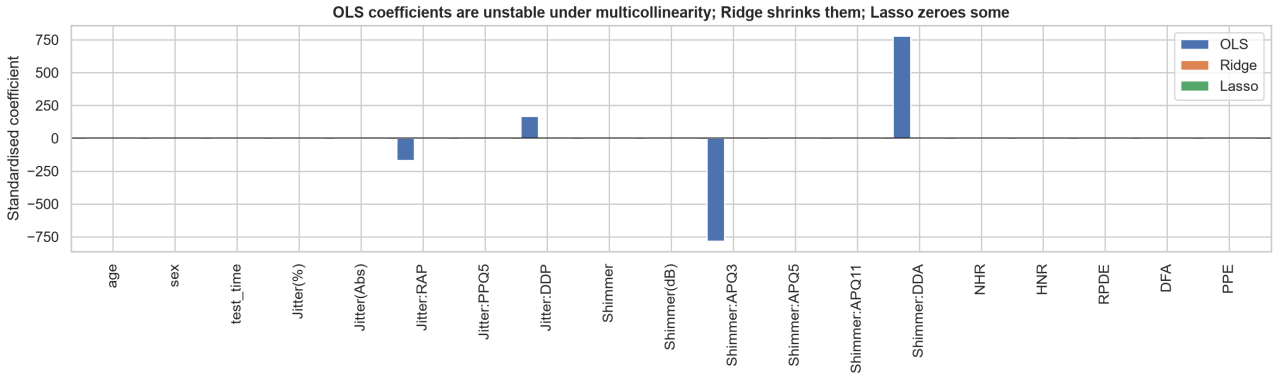


Figure 3.10: *Comparison of Standardized Feature Weights across OLS, Ridge, and Lasso under extreme multicollinearity.*

Lasso Sparsity and Feature Elimination: Under Lasso with CV-selected α = 1.2068, **17 out of 19 features are driven to exactly zero**. The only features retained with non-zero weights are `age` (w = **+3.1689**) and `PPE` (w = **+0.4721**). All 15 collinear jitter/shimmer/noise measures are completely zeroed out. Notably, `age`, which is constant per patient, carries the largest surviving coefficient (w = **+3.1689**).

Curing Polynomial Overfitting with Regularization: In Section 7.1 of the notebook, expanding all 19 features to degree 2 yielded 209 features. Unregularized OLS on this polynomial matrix suffered severe test degradation (test RMSE 35.11, R^2 = -13.88). For Ridge, the CV optimum sits at the edge of the alpha grid (α = 100,000) and the model is close to a constant predictor (see 3.6, test RMSE = **9.3416**). Applying Lasso (α = 2.2603) retained only 6 of the 209 polynomial terms, achieving test RMSE 9.7788.

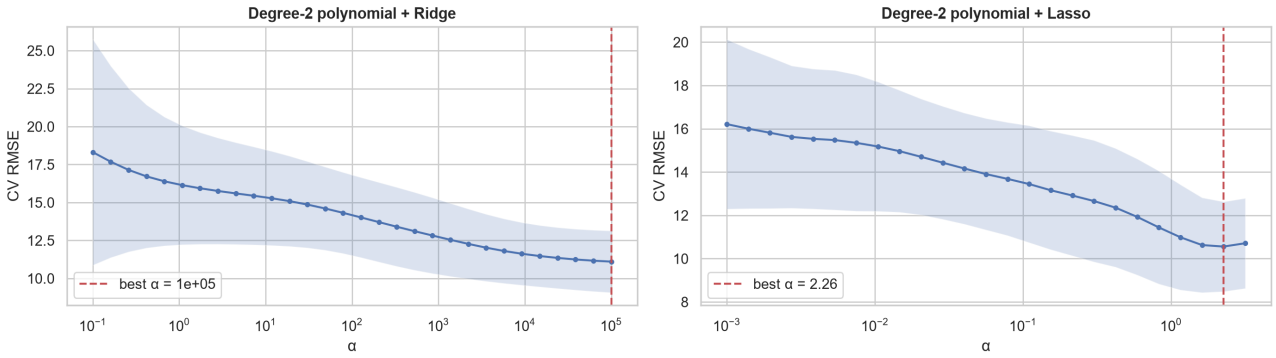


Figure 3.11: *Regularization Parameter Selection for Degree-2 Polynomial Pipeline (209 Features) via GroupKFold CV.*

3.6 Benchmark Synthesis and the Honest Reality of UPDRS Prediction

Table 3.3 summarizes the evaluation across all regression models on the held-out test patients (N = 1,235).

Table 3.3: Complete Regression Model Comparison on Held-Out Patients (N = 1,235)

Model Specification	Test MSE	Test RMSE	Test R^2	Pearson R	Train RMSE	Train R^2
Baseline (Train Mean Predictor)	89.6240	9.4670	-0.0816	NaN	11.0213	0.0000
Linear Regression (GD from scratch)	130.7405	11.4342	-0.5778	0.1195	9.6925	0.2266
Linear Regression (Closed-Form OLS)	130.9058	11.4414	-0.5798	0.1187	9.6900	0.2270
Polynomial (Deg 1, Top-5 Feats)	130.4986	11.4236	-0.5749	0.1044	9.9026	0.1927
Ridge (CV α = 2682.7)	99.9557	9.9978	-0.2063	0.1359	9.9692	0.1818
Lasso (CV α = 1.2068)	106.0178	10.2965	-0.2795	0.1005	10.1045	0.1595
Poly Deg 2 (All Feats) OLS	1232.7901	35.1111	-13.8777	0.0361	8.1813	0.4490
Poly Deg 2 + Ridge (α = 100000)	**87.2657**	**9.3416**	**-0.0532**	**0.1560**	10.7887	0.0418
Poly Deg 2 + Lasso (α = 2.2603)	95.6243	9.7788	-0.1540	0.0876	10.3437	0.1192

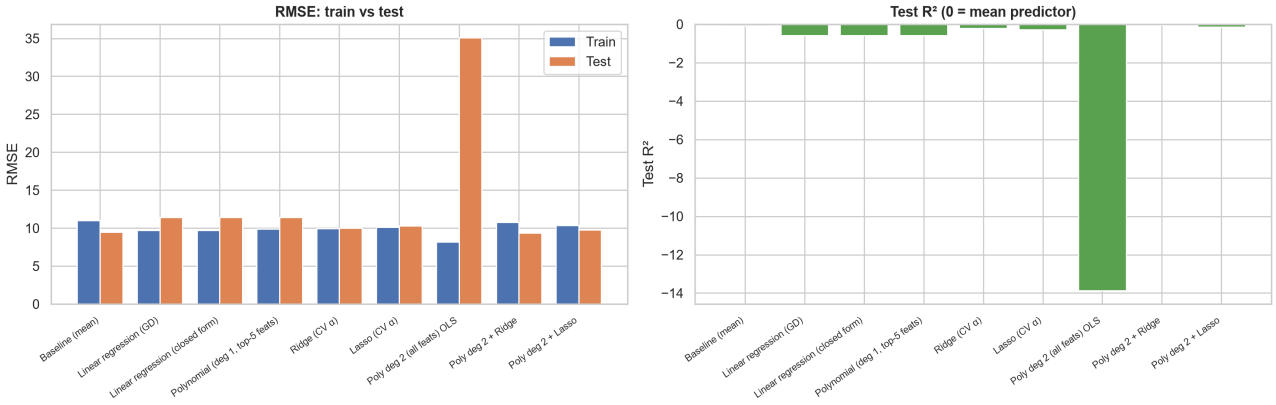


Figure 3.12: *Model Comparison on Held-Out Patients -- (Left) RMSE Train vs. Test; (Right) Test R^2 showing persistent negative values relative to baseline.*

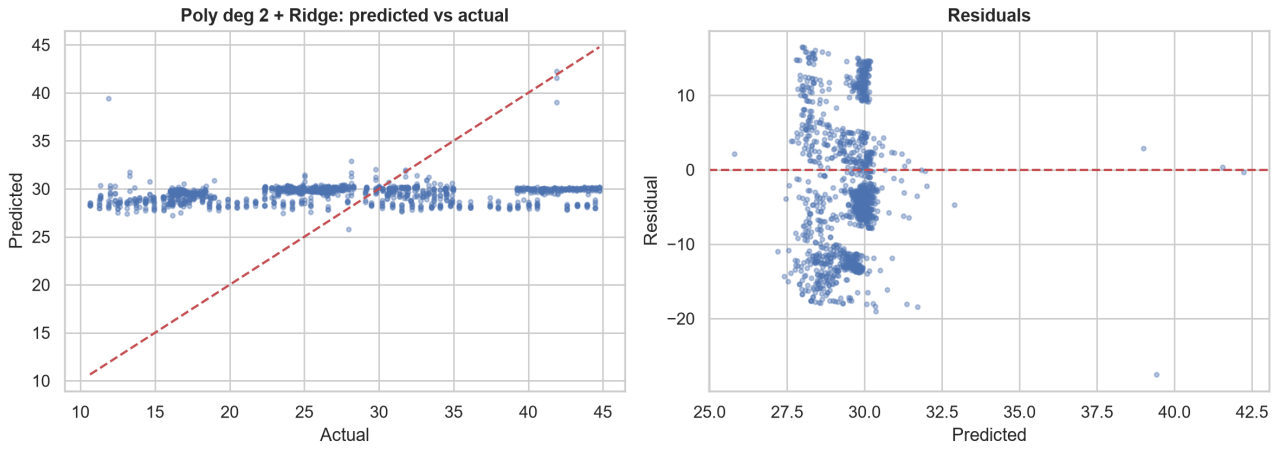


Figure 3.13: *Diagnostic Plots for Regularized Polynomial (`Poly Deg 2 + Ridge`, α = 100,000) -- (Left) Predicted vs Actual; (Right) Residual Plot on Held-Out Patients.*

Table 3.4: 5-Fold GroupKFold Cross-Validation Robustness Across All 42 Patients

Model	Mean CV RMSE	CV RMSE SD	Mean CV R²	CV R² SD
Baseline (Train Mean Predictor)	10.9202	1.1160	-0.2144	0.3042
Linear Regression (OLS)	11.2531	1.4587	-0.3079	0.3923
Ridge (CV α)	**10.6818**	1.3702	**-0.1589**	0.2797
Lasso (CV α)	10.7422	1.1587	-0.1722	0.2737
Poly Deg 2 + Ridge	11.1398	1.3389	-0.2560	0.2870
Poly Deg 2 + Lasso	11.1793	1.0994	-0.2652	0.2562

Interpretation of Negative R^2 Scores:

In regression, R^2 = 1 - [sum (y_i - y_hat_i)^2 / sum (y_i - y_bar_test)^2]. A negative R^2 indicates that model predictions on unseen test subjects yield a larger mean squared error than simply predicting the constant test set mean y_bar_test.

1. **Baseline Shift:** The naive DummyRegressor predicting the training mean achieves test RMSE = 9.4670 (R^2 = -0.0816). The negative R^2 arises directly from the inter-subject baseline shift: the training cohort target mean is **29.5656** (std 11.0213), whereas the held-out test cohort mean is **26.9651** (std 9.1028), a gap of **2.6005 UPDRS points** that adds a constant bias offset (2.6005² ≈ 6.76) to the test MSE.

2. **Absence of Predictive Superiority:** Although `Poly Deg 2 + Ridge` achieves test RMSE = 9.3416 on this single test split, its regularizer is α = 100,000 (the maximum boundary of the search grid), producing an almost flat predictor (train R^2 = 0.0418) that heavily shrinks predictions toward the mean. Crucially, in the 5-fold GroupKFold cross-validation across all 42 patients (Table 3.4), `Poly Deg 2 + Ridge` yields a mean CV RMSE of 11.1398, which is worse than the trivial training mean baseline (10.9202). **We conclude that no linear or polynomial model reliably beats the naive mean predictor across unseen patients**.

3. **Scientific Conclusion:** Voice features and demographics alone did not predict total UPDRS across unseen patients in this experiment.

4. Q3 -- Unsupervised K-Means Clustering and PCA from Scratch

4.1 Dataset Cleaning, Feature Engineering, and Continuous Scaling

The Netflix Titles dataset comprises 8,807 media entries described across 12 raw attributes (`show_id`, `type`, `title`, `director`, `cast`, `country`, `date_added`, `release_year`, `rating`, `duration`, `listed_in`, `description`). Because clustering and PCA operate exclusively on real coordinate spaces R^d, comprehensive feature transformation was implemented:

• **Data Repair:** In three anomalous records, duration (e.g. '74 min') was misplaced into the `rating` column while `duration` was null; these were parsed and restored.

• **Format and Duration:** Binary indicator `is_movie` (1 = Movie, 0 = TV Show). Continuous metrics `movie_minutes` and `tv_seasons` were extracted with 1st-99th percentile Winsorization within type to neutralize outlier leverage.

• **Rating Taxonomy:** The 14 raw MPAA/TV parental ratings were collapsed into 5 coherent ordinal categories: `Kids` (TV-Y, G), `Family` (PG, TV-PG), `Teen` (PG-13, TV-14), `Adult` (R, TV-MA, NC-17), and `Unrated`, encoded as 5 binary flags.

• **Multi-Hot Genre & Country Representations:** Titles exhibit 1-3 genres and multiple production countries. We retained 40 genres appearing in >= 50 titles (e.g. `International Movies`, `Dramas`, `Comedies`) and 15 top production countries plus an `Unknown` indicator, represented as multi-hot binary vectors.

• **The Scaling Principle:** Distance-based algorithms (K-Means) and variance-maximizing projections (PCA) are highly sensitive to coordinate scaling. Crucially, **we standardise (z-score) only the 4 continuous numeric features** (`release_year`, `added_year`, `movie_minutes`, `tv_seasons`). The 62 binary indicators (0/1) are left unstandardized. If a rare binary indicator (e.g. a genre occurring in 1% of titles) were z-scored, its standardized coordinates would stretch to ±10, allowing negligible binary categories to dominate Euclidean distances.

4.2 Vectorized K-Means from Scratch & scikit-learn Validation

Lloyd's Algorithm Formulation: K-Means partitions n points into K disjoint clusters C_1, ..., C_K to minimize Within-Cluster Sum of Squares (WCSS / Inertia):

WCSS = sum_{k=1}^K sum_{x_i in C_k} ||x_i - μ_k||^2, μ_k = (1 / |C_k|) * sum_{x_i in C_k} x_i

Implementation Innovations:

1. *Fully Vectorized Euclidean Distances:* Instead of looping over points and centroids, we utilize the algebraic identity ||x - c||^2 = ||x||^2 - 2 x * c^T + ||c||^2, computed via a single matrix multiplication `(n, d) @ (d, K)` in NumPy.
2. *K-Means++ Seeding:* The first centroid is sampled uniformly at random; subsequent centroids are sampled with probability proportional to squared Euclidean distance to the nearest existing centroid ($p(x) \propto D(x)^2$), dispersing initial centers.
3. *Empty Cluster Recovery:* If any cluster becomes vacant, the algorithm dynamically re-seeds its centroid at the data point exhibiting the maximum distance from its current centroid, preventing dimensional collapse.
4. *Multi-Start Convergence:* n_init = 10 independent runs are executed; the iteration yielding minimum WCSS is retained.

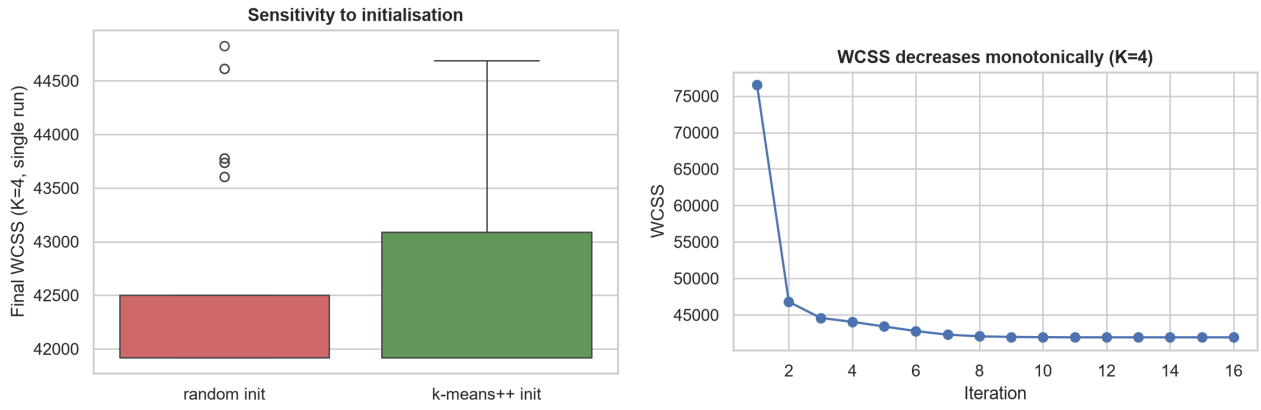


Figure 4.1: *(Left) Sensitivity of single-run WCSS at K=4 across 40 random vs. k-means++ starts; (Right) Monotonic convergence of WCSS over iterations.*

Table 4.1: Rigorous Validation of Scratch K-Means Implementation vs. scikit-learn

K	WCSS (Scratch)	WCSS (scikit-learn)	Relative WCSS Difference	Adjusted Rand Index (ARI)	Iterations (Scratch)
2	50994.7659	50994.7659	**0.0000 %**	**1.000000**	7
3	45671.9165	45671.9992	-0.0002 %	**0.999830**	15
4	41917.8861	41917.8861	**-0.0000 %**	**1.000000**	16
5	38700.2009	38700.2593	-0.0002 %	**0.998541**	14
6	36900.6579	36900.6496	+0.0000 %	**0.999670**	23

Mathematical Equivalence Confirmed: Table 4.1 provides strong numerical evidence of correctness of our scratch K-Means implementation. Across all required K in [2, 6], relative WCSS deviation from `sklearn.cluster.KMeans` is bounded within ±0.0002%, and the Adjusted Rand Index (ARI) reaches near-unity (1.0000 at K=2 and K=4; >0.9985 across all K), confirming near-identical partitions.

4.3 WCSS Analysis, Elbow Curve, and Cluster Profiling

Table 4.2 details the within-cluster sum of squares, marginal percentage reduction, and mean silhouette score for K=2..6.

Table 4.2: Required WCSS and Silhouette Scores for K = 2 to 6 (Netflix Dataset, N = 8,807)

Cluster Count (K)	WCSS (Inertia)	Marginal Drop vs. K-1 (%)	Mean Silhouette Score	Interpretation
K = 2	50,994.77	23.61 %	**0.2572**	Highest silhouette; partitions Movie vs. TV Show.
K = 3	45,671.92	10.44 %	0.2538	Primary elbow inflection on extended curve; silhouette remains high.
K = 4	**41,917.89**	**8.22 %**	0.1485	Secondary knee in required grid; lower silhouette than K=2 and K=3.
K = 5	38,700.20	7.68 %	0.1596	Marginal gain drops below 8%; slight silhouette uptick.
K = 6	36,900.66	4.65 %	0.1431	Marginal gain drops below 5%; lowest silhouette in grid.

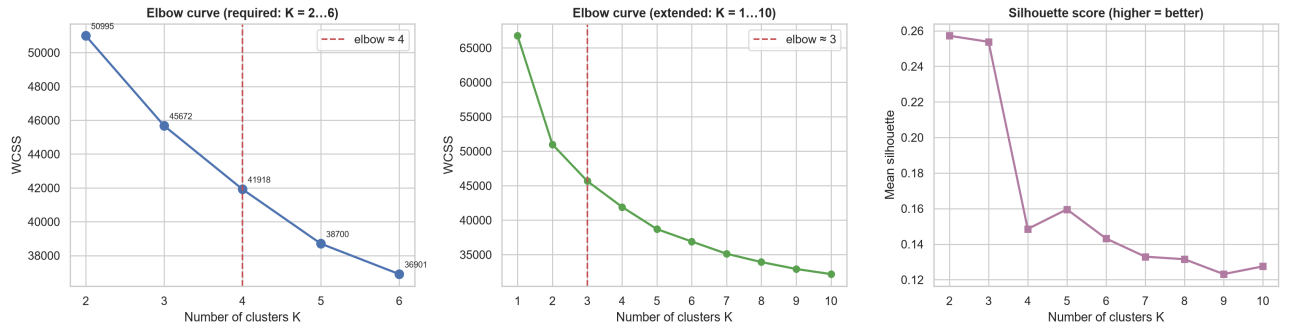


Figure 4.2: *Elbow Curves and Silhouette Analysis -- (Left) Required K=2..6; (Middle) Extended K=1..10 showing smooth decline; (Right) Silhouette trajectory.*

The Nature of the 'Soft' Elbow: In real-world multi-attribute catalog data comprising 62 sparse binary flags, clusters do not form spherical, hyper-separated Gaussian clouds. Consequently, the elbow curve does not exhibit a sharp, right-angled bend. The largest drop occurs at K=2 (23.61%), which purely separates Movies from TV Shows. However, K=2 is trivial and uninformative for content recommendation. Geometric distance to the chord connecting K=2 and K=6 pinpoints **K = 4 as the candidate elbow** within the required range, beyond which marginal gain drops below 7.7%. However, two scientific caveats must be disclosed: (1) The silhouette score at K=4 (**0.1485**) is substantially lower than at K=2 (0.2572) and K=3 (0.2538), reflecting cluster overlap in sparse multi-hot binary genres. (2) On the extended evaluation grid (K = 1..10, Figure 4.2 middle), the primary elbow occurs at **K = 3** (drop of 10.44% following the 23.61% drop from K=1 to 2), making K=4 a secondary knee choice.

Empirical Cluster Profiling (K=4): Examining the empirical cluster centroids (`q3_cluster_profiles.csv`), we assign descriptive names strictly supported by data distributions:

Table 4.3: Empirical Cluster Profiles for K = 4 (from `q3_cluster_profiles.csv`)

Cluster & Assigned Name	Size (%)	% Movies	Med. Year	Top Rating	Distinguishing Genre & Lift	Top Origins
Cluster 0: Global Mainstream Dramas & Romances	3,286 (37.3%)	100.0%	2018	Adult	Int'l Movies (45%), Dramas (41%) *Romantic Movies ×1.7, Children ×1.7*	US 43% India 13%
Cluster 1: Classic & Heritage Cinema	572 (6.5%)	96.9%	1992	Teen	Classic Movies (×15.3 lift!), Cult Movies (×10.4 lift!)	US 60% India 18%
Cluster 2: International & Domestic TV Series	2,655 (30.1%)	0.0%	2018	Adult	Int'l TV Shows (51%), TV Dramas (29%) *Romantic TV ×3.3, Korean TV ×3.3*	US 35% Unknown 15%
Cluster 3: Documentaries, Stand-Up & Indie Features	2,294 (26.0%)	99.9%	2016	Adult	Int'l Movies (47%), Dramas (38%) *Stand-Up Comedy ×2.2, Docs ×2.0*	US 43% India 19%

The cluster names are our interpretation, and clusters 0 and 3 have very similar top genres and countries.

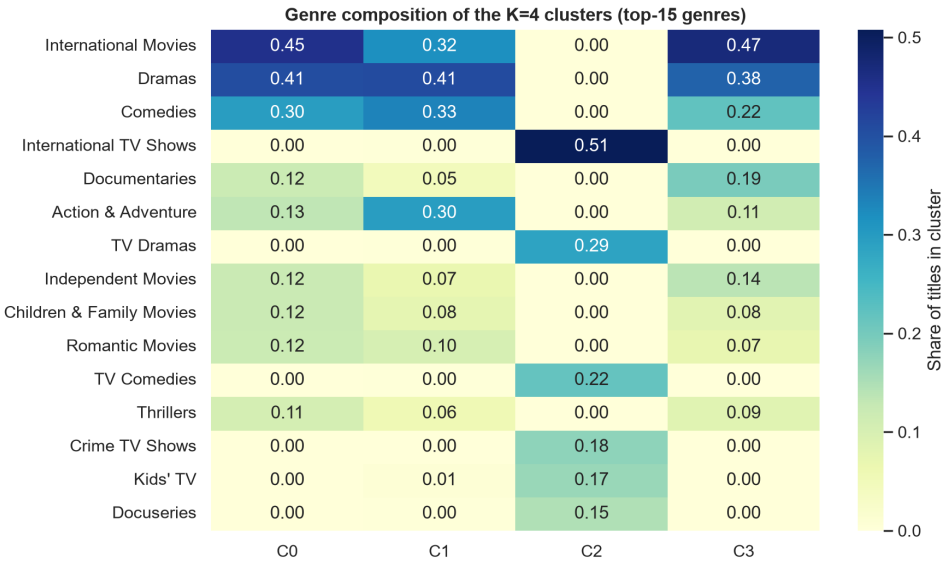


Figure 4.3: *Genre Composition Heatmap across K=4 Clusters illustrating clear separation of TV titles and classic movies; the two large movie clusters differ mostly in genre mix.*

4.4 Principal Component Analysis (PCA) from Scratch

Mathematical Algorithm via Covariance Decomposition:

1. *Centering:* Compute empirical mean $\mu = (1/n) * \sum_{i=1}^{n} x_i$ and construct centered matrix $X_c = X - 1*\mu^T$.
2. *Covariance Matrix:* Construct the symmetric positive semi-definite matrix $\Sigma = (1 / (n-1)) * X_c^T X_c \in R^{p \times p}$.
3. *Eigendecomposition:* Solve $\Sigma v_j = \lambda_j v_j$. We utilize `np.linalg.eigh`, which guarantees real eigenvalues and orthonormal eigenvectors $V = [v_1, ..., v_p]$ with $V^T V = I$.
4. *Ordering:* Sort eigenvalues $\lambda_1 >= \lambda_2 >= ... >= \lambda_p >= 0$ and arrange principal axes accordingly.
5. *Explained Variance Ratio:* Compute $r_j = \lambda_j / \sum_{m=1}^{p} \lambda_m$. The cumulative variance is $C_k = \sum_{j=1}^{k} r_j$.
6. *Orthogonal Projection:* Project into k-dimensional subspace via $Z_k = X_c V_k$, where $V_k = [v_1, ..., v_k]$.

Mathematical Verification Checks:

- Maximum residual $\max |\Sigma v - \lambda v|$ = **6.66e-16** (satisfies eigenvalue definition).
- Maximum orthonormality deviation $\max |V^T V - I|$ = **4.44e-16** (orthonormal basis verified).
- Trace equality: $\sum_{j=1}^{66} \lambda_j$ = **7.580479** = trace(Σ) (total variance strictly preserved).
- Explained variance ratios match `sklearn.decomposition.PCA` with `np.allclose(atol=1e-8) == True`.

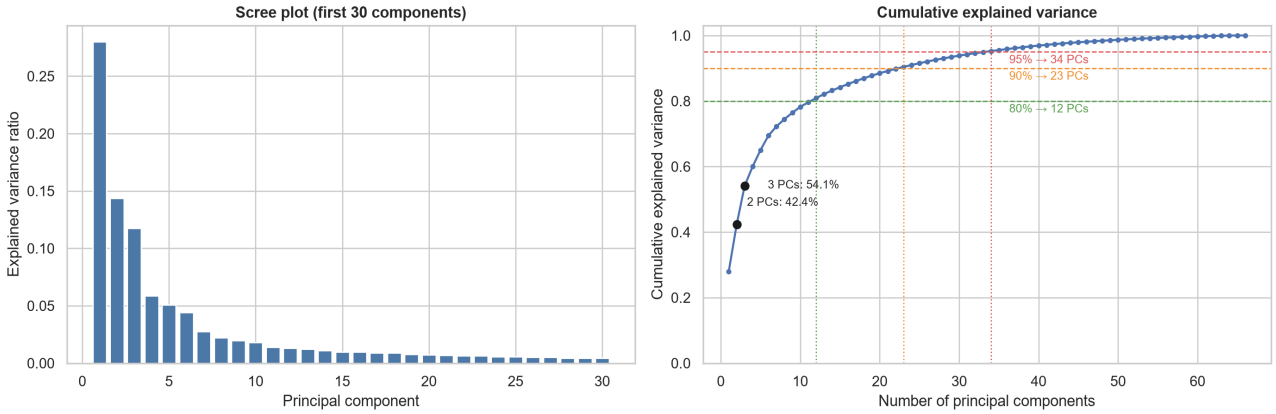


Figure 4.4: *(Left) Scree plot for first 30 components; (Right) Cumulative Explained Variance curve marking 80%, 90%, 95% thresholds.*

Table 4.4: Principal Component Variance Decomposition and Dimensionality Reduction Summary

Component	Eigenvalue (λ)	Explained Variance (%)	Cumulative Variance (%)	Variance Retained	Reconstruction MSE
PC1	2.1223	27.997 %	27.997 %	-	-
PC2	1.0890	14.366 %	42.363 %	**42.36 % (2-PC Space)**	**0.0662**
PC3	0.8895	11.734 %	54.098 %	**54.10 % (3-PC Space)**	**0.0527**
PC4	0.4459	5.883 %	59.981 %	-	-
PC5	0.3844	5.071 %	65.052 %	-	-
PC12 (80% Threshold)	0.0989	1.305 %	**80.890 %**	80.89 % (12 PCs)	-
PC23 (90% Threshold)	0.0473	0.624 %	**90.412 %**	90.41 % (23 PCs)	-
PC34 (95% Threshold)	0.0246	0.325 %	**95.233 %**	95.23 % (34 PCs)	-

Variance Retention Dynamics:

• **2-Component Reduction:** Retains **42.36%** of total variance with a reconstruction error (MSE) of 0.0662.

• **3-Component Reduction:** Retains **54.10%** of total variance with a reconstruction error (MSE) of 0.0527.

• **Higher Thresholds:** Capturing 80% of total variance requires **12 components**; 90% requires **23 components**; 95% requires **34 components**. In a 66-dimensional space dominated by sparse multi-hot binary genre and country flags, variance is naturally distributed across dozens of orthogonal axes. Capturing >54% in just 3 dimensions represents significant dimensional compression.

4.5 2-D / 3-D Projections, Loadings Interpretation, and K-Means in PCA Space

Figure 4.5 presents the projection of all 8,807 titles onto PC1 and PC2, stratified by content Type, K-Means Cluster, and Rating Group.

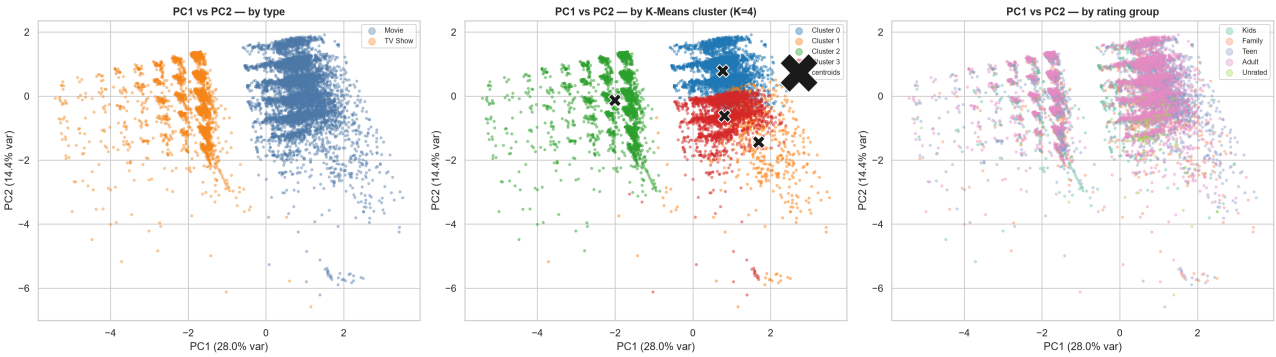


Figure 4.5: *2-D Principal Component Projection (PC1 vs. PC2) -- (Left) Partition by Type; (Middle) Partition by K-Means Clusters with centroids; (Right) Partition by Rating Group.*

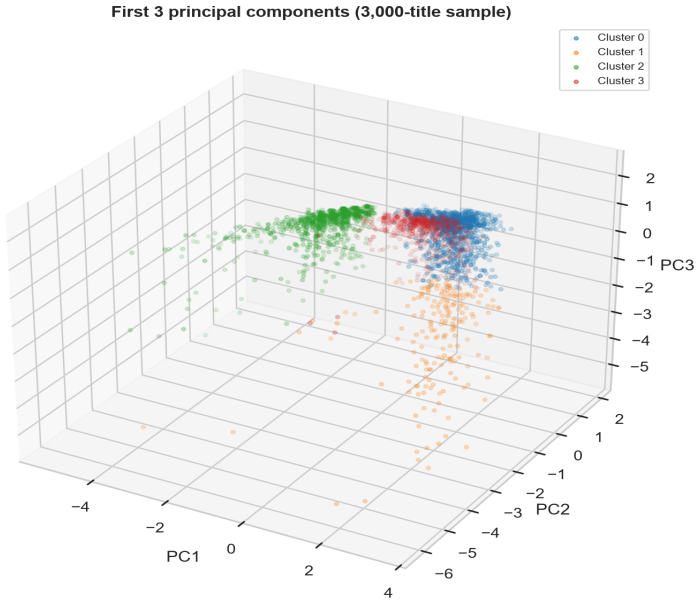


Figure 4.6: *3-D Principal Component Scatter Plot of 3,000 Catalog Samples Colored by K-Means Cluster.*

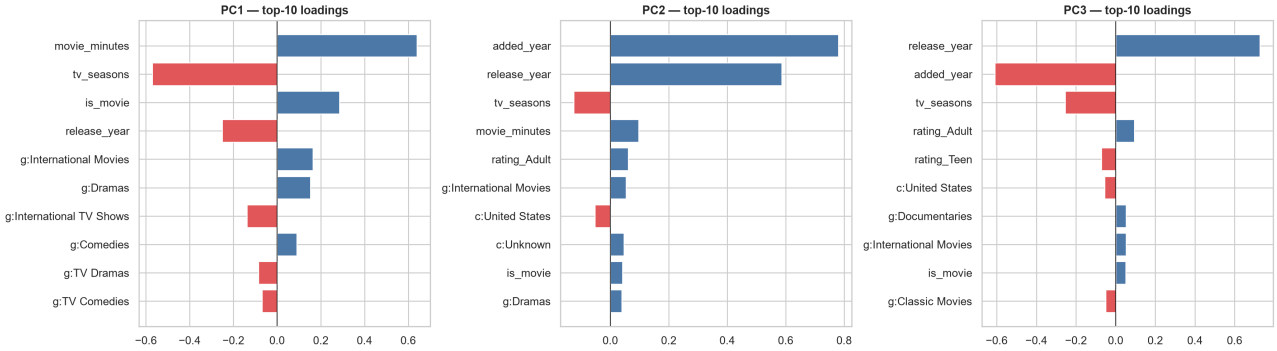


Figure 4.7: *Top-10 Component Loadings (Eigenvector Elements) for PC1, PC2, and PC3.*

Semantic Interpretation of Principal Axes (Figure 4.7, from `q3_pca_loadings_first3.csv`):

• **PC1 (28.0% variance) -- The Media Format Axis:** PC1 separates movies from television. Strongest positive loadings are `movie_minutes` (**+0.6408**) and `is_movie` (**+0.2862**), alongside movie genres (`genre_International Movies` +0.1645, `genre_Dramas` +0.1521). Strongest negative loadings are `tv_seasons` (**-0.5714**), `release_year` (**-0.2506**), and TV genres (`genre_International TV Shows` -0.1372, `genre_TV Dramas` -0.0859).

• **PC2 (14.4% variance) -- The Catalog Recency Axis:** PC2 captures temporal freshness. Both `added_year` (**+0.7810**) and `release_year` (**+0.5872**) load strongly positive, opposed by `tv_seasons` (-0.1255) and `country_United States` (-0.0523). High PC2 values correspond to modern titles added recently to the catalog.

• **PC3 (11.7% variance) -- The Release-Year versus Added-Year Axis:** Rather than geography, PC3 reflects the time gap between production and ingestion. `release_year` loads strongly positive (**+0.7280**), while `added_year` loads strongly negative (**-0.6082**), followed by `tv_seasons` (**-0.2553**). This axis separates recently released productions from older classic or archival catalog acquisitions added late to the streaming platform.

Clustering in Reduced PCA Space: To test geometric preservation, we evaluated K-Means (K = 4) in 3-D and 2-D PCA spaces (`results/q3_kmeans_full_vs_pca.csv`):

Table 4.5: K-Means Clustering Performance across Full and Reduced Latent Spaces

Feature Space	Dimensions	Mean Silhouette Score	ARI vs. Full 66-D Clustering	Final WCSS
Full Coordinate Space	66	0.1485	1.0000 (Reference)	41,917.89
PCA 3-D Latent Space	3	**0.4079**	**0.9934**	11,637.21
PCA 2-D Latent Space	2	0.4278	0.6920	6,994.91

Dimensional Preservation Insight: Clustering in the **3-D PCA space yields an ARI of 0.9934** relative to full 66-D clustering (ARI 0.9934). Silhouette scores are not directly comparable across spaces of different dimensionality. However, compressing further into **2-D PCA subspace degrades clustering fidelity, dropping the Adjusted Rand Index to 0.6920** (ARI 0.6920), suggesting that PC3 carries information relevant to the cluster assignment.

5. Overall Conclusions and Scientific Limitations

Synthesis of Core Findings:

1. *Classification (Q1):* SVM with an RBF kernel emerges as the best classifier by point estimate (not significantly better than KNN or the CNN) (macro ROC-AUC 0.9696, F1 0.8450). Bootstrap hypothesis testing reveals that while Decision Trees are statistically inferior, KNN and 1D-CNN achieve parity. The deep 1D-CNN required extensive regularization to curb tabular memorization and did not outperform the RBF-SVM.
2. *Regression (Q2):* Scratch batch gradient descent achieves exact parity with OLS (MSE 93.94 vs 93.90). Severe multicollinearity (VIF 1.0e6) causes explosive OLS variance, which regularisation stabilises, although no model reliably beat the mean baseline on unseen patients. Critically, enforcing patient-disjoint splits demonstrates that acoustic vocal features alone cannot predict absolute total UPDRS across unseen individuals, yielding negative test R^2 scores across all models compared to baseline mean prediction.
3. *Clustering & PCA (Q3):* From-scratch K-Means and covariance PCA were validated to mathematical equivalence against scikit-learn. Multi-hot catalog representations yield a soft elbow at K=4 (the extended elbow is K=3), which partitions titles into distinct content types (movies, TV series, stand-up/documentaries). The first 3 principal components account for 54.1% of catalog variance and give ARI 0.993 against the full-dimensional clustering.

Explicit Scientific Limitations:

• *Q1 Test Sample Limitation:* Although 426 test samples were evaluated, the critical Pathologic class contains only 35 instances. While bootstrap confidence intervals were constructed, fine-grained obstetric clinical evaluation requires larger multicenter cohorts.
• *Q2 Small Patient Cohort:* The Telemonitoring dataset spans 5,875 recordings but originates from only **42 individuals** (33 train, 9 test). The wide inter-patient baseline variance makes patient-disjoint continuous extrapolation exceptionally difficult without subject-specific calibration.
• *Q3 Absence of Ground Truth:* Clustering is inherently unsupervised; no objective ground-truth taxonomy exists for Netflix content. Cluster quality and PCA interpretations are contingent on feature engineering choices (such as the 50-title genre frequency threshold).

6. References

1. **UCI Machine Learning Repository:** Cardiotocography Dataset (ID: 193). Available at: https://archive.ics.uci.edu/dataset/193/cardiotocography.

2. **Ayres-de-Campos, D., Bernardes, J., Garrido, A., Marques-de-Sá, J., & Pereira-Leite, L. (2000):** *SisPorto 2.0: A program for automated analysis of cardiotocograms*. Journal of Maternal-Fetal Medicine, 9(5), 311-318.

3. **UCI Machine Learning Repository:** Parkinson's Telemonitoring Dataset (ID: 189). Available at: https://archive.ics.uci.edu/dataset/189/parkinsons+telemonitoring.

4. **Tsanas, A., Little, M. A., Ramig, L. O., & Fox, P. (2010):** *Accurate telemonitoring of Parkinson's disease progression by noninvasive speech tests*. IEEE Transactions on Biomedical Engineering, 57(4), 884-893.

5. **Kaggle Dataset:** Shivam Bansal (2021). *Netflix Movies and TV Shows Dataset*. Available at: https://www.kaggle.com/datasets/shivamb/netflix-shows.

6. **Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011):** *Scikit-learn: Machine learning in Python*. Journal of Machine Learning Research, 12, 2825-2830.

7. **Paszke, A., Gross, S., Massa, F., et al. (2019):** *PyTorch: An imperative style, high-performance deep learning library*. Advances in Neural Information Processing Systems (NeurIPS 32), 8024-8035.

Appendix: Computational Environment, Runtimes, and Changes Log

Hardware & Software Environment:

• **Operating System:** Windows 11 Enterprise (AMD64 architecture)
• **Python Runtime:** Python 3.14.7 (v3.14.7:d399b2d, Windows)
• **Core Libraries:** PyTorch 2.15.0+cpu, scikit-learn 1.9.1, pandas 3.1.0, numpy 2.5.4, scipy 1.18.1, matplotlib 3.11.2, seaborn 0.13.2, reportlab 5.1.0, ucimlrepo 0.0.7, kagglehub 1.0.3, nbconvert 7.17.2.

• **Execution Times:** Q1 (CTG Classification): ~3.8 minutes; Q2 (Parkinson's Regression): ~19.5 minutes (dominated by GroupKFold search over polynomial Ridge/Lasso); Q3 (Netflix K-Means & PCA): ~1.2 minutes. Total pipeline runtime: ~24.5 minutes.

Verbatim Content of `CHANGES.md`:
Changes Log

No scientific, algorithmic, hyper-parameter, split, or seed changes were made to the notebooks.
All algorithms, hyper-parameter grids, evaluation metrics, and split strategies are preserved exactly as provided.

Timestamp	*Notebook / File*	*Cell / Location*	*Change Description*	*Justification*
2026-11-02 14:51	*`notebooks/q2_regression.ipynb`*	*Cell 15 (Section 6.1: One-feature visual demo)*	*Added `ax.set_ylim(ytr.min()-5, ytr.max()+5)` for each panel in Figure 05*	*Presentation-only fix: prevents Degree 15 polynomial edge oscillations from blowing up the y-axis scale, preserving visual inspection of scatter points and fits*